

Aula 5

Monday, March 16, 2026 10:05 AM

Multiclass Classification

- (1) $\rightarrow$ Multiple binary classifier
- (2) $\rightarrow$ Multinomial logistic Regression

Dataset

$$D = \{ (x^{(i)}, y^{(i)}) : x^{(i)} \in \mathbb{R}^d, y^{(i)} \in \{1, 2, \dots, K\}, i=1, \dots, N \}$$

estimate $P(y=k|x)$

(1) Multiple binary classifier

1.1

OVA $\rightarrow$ One Versus all
 $\hookrightarrow$ para cada K
 Siga $h_K(x) = \sigma(w_K \tilde{x})$
 $\hat{y} = \arg \max_K \{ \hat{p}_K \}$
 $\hat{p}_K = h_K(x)$

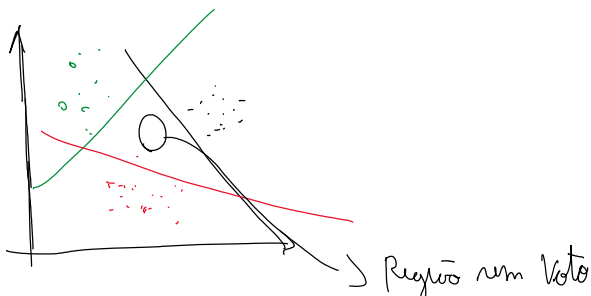
1.2 $\hookrightarrow O(n)$

OVO $\rightarrow$ One Versus one
 $\hookrightarrow$ combinations
 $\hookrightarrow \forall k_1, k_2$

$$\hat{y} = \arg \max_K \left\{ \sum_{j \neq k} h_{kj}(x) \right\}$$

 $\hookrightarrow O(n^2)$

Classifier response binary



(2) Logistic Multinomial Regression

- $\hookrightarrow$ hipótese $P(y=1|x)$
- $\hookrightarrow P(y=0|x) = 1 - P(y=1|x)$

$$\hat{p}_1 = P(y=1|x)$$

$$\hat{p}_0 = 1 - \hat{p}_1$$

$$\hat{p}_1 = \sigma(w^T \tilde{x}) = \frac{1}{1 + e^{-w^T \tilde{x}}} = \frac{e^{w^T \tilde{x}}}{1 + e^{w^T \tilde{x}}}$$

$$\hat{p}_0 = 1 - \hat{p}_1 = \frac{1}{e^{w^T \tilde{x}} + 1}$$

Since w_0 , define $w_L = w + w_0$

$$\hat{p}_0 = \dots$$

$$\hat{p}_1 = \dots$$

$$\hat{p}_k = \frac{e^{w_k \tilde{x}}}{\sum_{j=0}^K e^{w_j \tilde{x}}}$$

Loss Function

Before

$$L(w) = -\frac{1}{N} \sum_{n=1}^N y^{(n)} \ln(\hat{p}^{(n)}) + (1 - y^{(n)}) \ln(1 - \hat{p}^{(n)})$$

Now

$$L(w) = -\frac{1}{N} \sum_{n=1}^N y_1^{(n)} \ln(\hat{p}_1^{(n)}) + y_0^{(n)} \ln(\hat{p}_0^{(n)})$$

Multinomial

$$\hat{p}_k = P(y=k|x) = \frac{e^{w_k \tilde{x}}}{\sum_{j=1}^K e^{w_j \tilde{x}}}$$

↳ softmax activation

Performance Metrics

	A	B	C
Predict \ class	A TP _{AA}	B TP _{AB}	C TP _{AC}
	B FP _{BA}	TP _{BB}	
	C FP _{CA}		TP _{CC}

Macro - Average

$$TP = \sum TP$$

$$FP = \sum FP_k$$

Polys - Any (many important / tasks as dense)

$$\text{Recall}_k = \frac{TP_k}{TP_k + FN_k}$$

$$\text{Recall} = \sum_k \text{Recall}_k$$

Linear and non-linear

Non Linear function

$$g(x) = w_0 + w_1 x_1^2 + w_2 x_1 x_2 + w_3 x_2^2$$

Generalized linear models

$$\beta_1 = x_1^2$$

$$\beta_2 = x_1 x_2$$

$$\beta_3 = x_2^2$$

Perceps

- ↳ Encuentra representos $z = \phi(x)$
- ↳ Percepsos computan g
- ↳ Percepsos en g pueden estar muy grande

Percepsions

